CS 440: Introduction to Artificial Intelligence
Assignment 3 — Probabilistic Reasoning

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1 Problem 1 (15 points): Bayesian Network

The Bayesian network contains five Boolean variables A through E. A, B, and C are root nodes; D depends on A and B; E depends on B and C.

$$P(A = T) = 0.2, \quad P(B = T) = 0.5, \quad P(C = T) = 0.8$$

A	B	$P(D = T \mid A, B)$
F	F	0.9
F	T	0.6
T	F	0.5
T	T	0.1

B	C	$P(E = T \mid B, C)$
F	F	0.2
F	T	0.4
T	F	0.8
T	T	0.3

The joint probability factorises as:

$$P(A, B, C, D, E) = P(A) \cdot P(B) \cdot P(C) \cdot P(D \mid A, B) \cdot P(E \mid B, C)$$

(a) All five variables simultaneously true

$$\begin{aligned} P(A, B, C, D, E) &= P(A) \cdot P(B) \cdot P(C) \cdot P(D = T \mid A = T, B = T) \cdot P(E = T \mid B = T, C = T) \\ &= 0.2 \times 0.5 \times 0.8 \times 0.1 \times 0.3 \\ &= \boxed{0.0024} \end{aligned}$$

(b) All five variables simultaneously false

Using $P(D = F \mid A = F, B = F) = 1 - 0.9 = 0.1$ and $P(E = F \mid B = F, C = F) = 1 - 0.2 = 0.8$:

$$\begin{aligned} P(\neg A, \neg B, \neg C, \neg D, \neg E) &= 0.8 \times 0.5 \times 0.2 \times 0.1 \times 0.8 \\ &= \boxed{0.0064} \end{aligned}$$

(c) $P(\neg A \mid B, C, D, E)$

Using the normalisation trick with $\alpha = \dfrac{1}{P(A, B, C, D, E) + P(\neg A, B, C, D, E)}$:

$$P(\neg A \mid B, C, D, E) = \frac{P(\neg A, B, C, D, E)}{P(A, B, C, D, E) + P(\neg A, B, C, D, E)}$$

The numerator uses $P(D = T \mid A = F, B = T) = 0.6$:

$$P(\neg A, B, C, D, E) = 0.8 \times 0.5 \times 0.8 \times 0.6 \times 0.3 = 0.0576$$

$$P(\neg A \mid B, C, D, E) = \frac{0.0576}{0.0024 + 0.0576} = \frac{0.0576}{0.0600} = \boxed{0.96}$$

Given that B, C, D, E are all true, there is a **96%** probability that A is false. The high prior $P(\neg A) = 0.8$, combined with the favourable $P(D = T \mid \neg A, B = T) = 0.6$, drives this result.

2 Problem 2 (15 points): Burglary Network

Network parameters:

$$P(B) = 0.001, \quad P(E) = 0.002$$

B	E	$P(A = T \mid B, E)$	$P(A = F \mid B, E)$
T	T	0.95	0.05
T	F	0.94	0.06
F	T	0.29	0.71
F	F	0.001	0.999

A	$P(J = T \mid A)$
T	0.90
F	0.05

A	$P(M = T \mid A)$
T	0.70
F	0.01

(a) P(**Burglary** $\mid J = T, M = T$)

With $\alpha = \frac{1}{P(J,M)}$, we marginalise over E and A:

$$P(B \mid J, M) = \alpha\, P(B) \sum_E P(E) \sum_A P(A \mid B, E)\, P(J = T \mid A)\, P(M = T \mid A)$$

Inner sum over A for each (B, E) pair:

B	E	$A = T$ term	$A = F$ term	$f(B, E)$
T	T	$0.95 \times 0.9 \times 0.7 = 0.598500$	$0.05 \times 0.05 \times 0.01 = 0.000025$	0.598525
T	F	$0.94 \times 0.9 \times 0.7 = 0.592200$	$0.06 \times 0.05 \times 0.01 = 0.000030$	0.592230
F	T	$0.29 \times 0.9 \times 0.7 = 0.182700$	$0.71 \times 0.05 \times 0.01 = 0.000355$	0.183055
F	F	$0.001 \times 0.9 \times 0.7 = 0.000630$	$0.999 \times 0.05 \times 0.01 = 0.000500$	0.001130

Marginalising over E:

$$\begin{aligned}
P(B = T, J, M) &= \alpha \times 0.001 \times \big[0.002 \times 0.598525 + 0.998 \times 0.592230\big] \\
&= \alpha \times 0.001 \times [0.001197 + 0.591046] \\
&= \alpha \times 0.001 \times 0.592243 \\
&= 0.000592\,\alpha
\end{aligned}$$

$$\begin{aligned}
P(B = F, J, M) &= \alpha \times 0.999 \times \big[0.002 \times 0.183055 + 0.998 \times 0.001130\big] \\
&= \alpha \times 0.999 \times [0.000366 + 0.001128] \\
&= \alpha \times 0.999 \times 0.001494 \\
&= 0.001493\,\alpha
\end{aligned}$$

Normalisation:

$$\alpha = \frac{1}{0.000592 + 0.001493} = \frac{1}{0.002085} \approx 479.6$$

$\boxed{P(\text{Burglary} = T \mid J = T, M = T) \approx 0.284}$ $P(\text{Burglary} = F \mid J = T, M = T) \approx 0.716$

This matches the result in the textbook.

(b) Chain network: enumeration vs. variable elimination

Consider a chain $X_1 \to X_2 \to \cdots \to X_n$ where we compute $P(X_1 \mid X_n = T)$.

Enumeration. We must sum over all 2^{n-2} joint assignments to $X_2, \ldots, X_{n-1}$. Each such assignment requires n multiplications (one per factor in the chain), so the total cost is

$$O\big(n \cdot 2^{n-2}\big) = O(n \cdot 2^n),$$

which is **exponential** in n.

More concretely, for n variables there are 2^{n-2} hidden-variable assignments, and each requires $(n-1)$ multiplications plus one for the evidence factor, giving approximately $n \cdot 2^{n-2}$ multiplications and $2^{n-2} - 1$ additions in the summation.

Variable elimination. We eliminate variables right-to-left. Starting from the rightmost hidden variable X_{n-1}:

$$P(X_1 \mid X_n = T) = \alpha\, P(X_1) \cdots \sum_{x_{n-2}} P(x_{n-2} \mid x_{n-3}) \underbrace{\sum_{x_{n-1}} P(x_{n-1} \mid x_{n-2})\, P(X_n = T \mid x_{n-1})}_{f_{X_{n-1}}(x_{n-2})}$$

Summing out X_{n-1} (a binary variable) costs:

- 2 multiplications (one per value of x_{n-1}),
- 1 addition, and
- produces a factor $f_{X_{n-1}}$ over the single variable x_{n-2}.

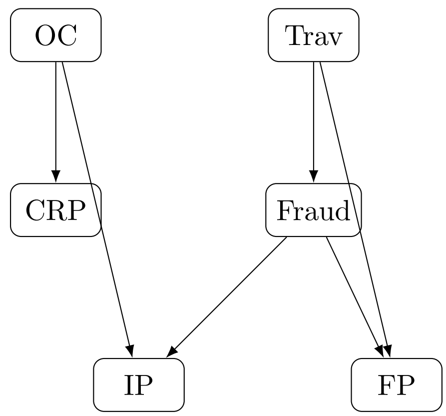
Each of the $n-2$ elimination steps has the same cost, so the total number of arithmetic operations is:

$$(n-2) \times (2 \text{ multiplications} + 1 \text{ addition}) \;+\; 1 \text{ normalisation division} = O(n),$$

which is **linear** in n — exponentially faster than enumeration.

3 Problem 3 (20 points): Fraud Detection

(a) Bayesian Network



Root nodes: *OC*, *Trav*. *Fraud* depends on *Trav*; *FP* depends on *Trav* and *Fraud*; *IP* depends on *Fraud* and *OC*; *CRP* depends on *OC*.

$$P(\text{Trav} = T) = 0.05, \qquad P(\text{OC} = T) = 0.75$$

Trav	$P(\text{Fraud} = T)$	$P(\text{Fraud} = F)$
T	0.010	0.990
F	0.004	0.996

Trav	Fraud	$P(\text{FP} = T)$	$P(\text{FP} = F)$
T	T	0.90	0.10
T	F	0.90	0.10
F	T	0.10	0.90
F	F	0.01	0.99

Note: When Trav$= T$, $P(\text{FP} = T) = 0.90$ regardless of Fraud, because "when the card holder is travelling, 90% of transactions are foreign purchases regardless of the legitimacy of the transaction."

Fraud	OC	$P(\text{IP} = T)$	$P(\text{IP} = F)$
T	T	0.020	0.980
T	F	0.011	0.989
F	T	0.010	0.990
F	F	0.001	0.999

OC	$P(\text{CRP} = T)$	$P(\text{CRP} = F)$
T	0.100	0.900
F	0.001	0.999

(b) Probability Queries

Query 1: $P(\textbf{Fraud})$

$$\begin{aligned}
P(\text{Fraud} = T) &= P(\text{Fraud} = T \mid \text{Trav} = T) \cdot P(\text{Trav} = T) + P(\text{Fraud} = T \mid \text{Trav} = F) \cdot P(\text{Trav} = F) \\
&= 0.01 \times 0.05 + 0.004 \times 0.95 \\
&= 0.0005 + 0.0038 = \boxed{0.0043}
\end{aligned}$$

Before any transaction evidence is observed, the baseline probability of fraud is **0.43%**.

Query 2: $P(\textbf{Fraud} \mid \textbf{FP} = T, \textbf{IP} = F, \textbf{CRP} = T)$ With $\alpha = \frac{1}{P(\text{FP}=T, \text{IP}=F, \text{CRP}=T)}$, marginalising over Trav and OC:

$$\begin{aligned}
P(\text{Fraud} \mid \text{FP}, \neg\text{IP}, \text{CRP}) = \alpha \sum_{\text{Trav}} & P(\text{Trav})\, P(\text{Fraud} \mid \text{Trav})\, P(\text{FP} = T \mid \text{Trav}, \text{Fraud}) \\
& \times \sum_{\text{OC}} P(\text{OC})\, P(\text{CRP} = T \mid \text{OC})\, P(\text{IP} = F \mid \text{Fraud}, \text{OC})
\end{aligned}$$

Inner OC sum for each value of Fraud:

$$\begin{aligned}
S_{\text{OC}}(\text{Fraud} = T) &= 0.75 \times 0.1 \times 0.980 + 0.25 \times 0.001 \times 0.989 \\
&= 0.07350 + 0.000247 = 0.073747 \\
S_{\text{OC}}(\text{Fraud} = F) &= 0.75 \times 0.1 \times 0.990 + 0.25 \times 0.001 \times 0.999 \\
&= 0.07425 + 0.000250 = 0.074500
\end{aligned}$$

Trav sum × OC sum for Fraud$= T$**:**

$$\Big[\underbrace{0.05 \times 0.01 \times 0.90}_{\text{Trav}=T} + \underbrace{0.95 \times 0.004 \times 0.10}_{\text{Trav}=F}\Big] \times 0.073747$$
$$= (0.000450 + 0.000380) \times 0.073747 = 0.000830 \times 0.073747 = 0.0000612\,\alpha^{-1}$$

Trav sum × OC sum for Fraud$= F$**:**

$$\Big[\underbrace{0.05 \times 0.99 \times 0.90}_{\text{Trav}=T} + \underbrace{0.95 \times 0.996 \times 0.01}_{\text{Trav}=F}\Big] \times 0.074500$$
$$= (0.044550 + 0.009462) \times 0.074500 = 0.054012 \times 0.074500 = 0.004024\,\alpha^{-1}$$

Normalisation:

$$\alpha = \frac{1}{0.0000612 + 0.004024} = \frac{1}{0.004085} \approx 244.8$$

$$\boxed{P(\text{Fraud} = T \mid \text{FP} = T, \text{IP} = F, \text{CRP} = T) \approx 0.015} \qquad P(\text{Fraud} = F \mid \cdots) \approx 0.985$$

The fraud probability increases from 0.43% to approximately **1.5%** given the evidence. The foreign purchase raises suspicion, but non-fraudulent travellers are also likely to make foreign purchases, keeping the posterior low.

4 Problem 4 (40 points): HMM Search and Rescue

The rover starts at $X_1 = A$. Positions $X_t \in \{A, B, C, D, E, F\}$; moves east with $p = 0.8$, stays with $p = 0.2$; F is absorbing. Volcanic vents at A and D give noiseless readings: $P(E_t = \text{hot} \mid X_t \in \{A, D\}) = 1$, $P(E_t = \text{cold} \mid X_t \notin \{A, D\}) = 1$.

Transition matrix $P(X_{t+1} \mid X_t)$:

	A	B	C	D	E	F
A	0.2	0.8	0	0	0	0
B	0	0.2	0.8	0	0	0
C	0	0	0.2	0.8	0	0
D	0	0	0	0.2	0.8	0
E	0	0	0	0	0.2	0.8
F	0	0	0	0	0	1

Observation model $P(E_t \mid X_t)$:

X_t	$P(\text{hot})$	$P(\text{cold})$
A	1	0
B	0	1
C	0	1
D	1	0
E	0	1
F	0	1

The forward-filtering recurrence is:

$$P(X_t \mid e_{1:t}) \propto P(E_t \mid X_t) \sum_{X_{t-1}} P(X_t \mid X_{t-1})\, P(X_{t-1} \mid e_{1:t-1})$$

Day 1. $X_1 = A$ with certainty; $E_1 = \text{hot}$ is consistent with A being a vent:

$$P(X_1 \mid E_1 = \text{hot}) = \langle 1, 0, 0, 0, 0, 0 \rangle$$

Day 2 (intermediate). Propagating forward from $X_1 = A$:

$$\sum_{X_1} P(X_2 \mid X_1)\, P(X_1 \mid e_1) = \langle 0.2, 0.8, 0, 0, 0, 0 \rangle$$

Multiplying element-wise by $P(E_2 = \text{cold} \mid X_2) = \langle 0, 1, 1, 0, 1, 1 \rangle$:

$$\propto \langle 0, 0.8, 0, 0, 0, 0 \rangle \xrightarrow{\times 1.25} \langle 0, 1, 0, 0, 0, 0 \rangle$$

$$P(X_2 \mid e_{1:2}) = \langle 0, 1, 0, 0, 0, 0 \rangle$$

1. Filtering: $P(X_3 \mid \mathbf{hot}_1, \mathbf{cold}_2, \mathbf{cold}_3)$

Propagating forward from $X_2 = B$ (certainty):

$$\sum_{X_2} P(X_3 \mid X_2)\, P(X_2 \mid e_{1:2}) = \langle 0, 0.2, 0.8, 0, 0, 0 \rangle$$

Multiplying element-wise by $P(E_3 = \text{cold} \mid X_3) = \langle 0, 1, 1, 0, 1, 1 \rangle$:

$$\propto \langle 0, 0.2, 0.8, 0, 0, 0 \rangle$$

This already sums to 1.0; no further normalisation needed.

X_3	$P(X_3 \mid \text{hot}_1, \text{cold}_2, \text{cold}_3)$
A	0.0
B	0.2
C	0.8
D	0.0
E	0.0
F	0.0

$$\boxed{P(X_3 \mid \text{hot}_1, \text{cold}_2, \text{cold}_3) = \langle 0, 0.2, 0.8, 0, 0, 0 \rangle}$$

After three days of observations (hot, cold, cold), the rover is most likely at position C with probability 0.8, or at B with probability 0.2.

2. Smoothing: $P(X_2 \mid \mathbf{hot}_1, \mathbf{cold}_2, \mathbf{cold}_3)$

Using the forward-backward algorithm:

$$P(X_2 \mid e_{1:3}) \propto P(X_2 \mid e_{1:2}) \;\odot\; \beta(X_2)$$

where the backward message is:

$$\beta(X_2) = P(E_3 = \text{cold} \mid X_2) = \sum_{X_3} P(X_3 \mid X_2)\, P(E_3 = \text{cold} \mid X_3)$$

X_2	Computation	$\beta(X_2)$
A	$0.2 \times P(\text{cold} \mid A) + 0.8 \times P(\text{cold} \mid B) = 0.2 \times 0 + 0.8 \times 1$	0.8
B	$0.2 \times 1 + 0.8 \times 1$	1.0
C	$0.2 \times 1 + 0.8 \times 0$	0.2
D	$0.2 \times 0 + 0.8 \times 1$	0.8
E	$0.2 \times 1 + 0.8 \times 1$	1.0
F	$0 \times 1 + 1 \times 1$	1.0

Backward message: $\beta = \langle 0.8, 1.0, 0.2, 0.8, 1.0, 1.0\rangle$.
Since $P(X_2 \mid e_{1:2}) = \langle 0, 1, 0, 0, 0, 0\rangle$ (nonzero only at B):

$$P(X_2 \mid e_{1:3}) \propto \langle 0.8, 1.0, 0.2, 0.8, 1.0, 1.0\rangle \odot \langle 0, 1, 0, 0, 0, 0\rangle = \langle 0, 1.0, 0, 0, 0, 0\rangle$$

X_2	$P(X_2 \mid \text{hot}_1, \text{cold}_2, \text{cold}_3)$
A	0.0
B	1.0
C	0.0
D	0.0
E	0.0
F	0.0

$$\boxed{P(X_2 \mid \text{hot}_1, \text{cold}_2, \text{cold}_3) = \langle 0, 1, 0, 0, 0, 0\rangle}$$

Using all three days of evidence, we can confirm with **certainty** that the rover was at position B on day 2.

4. Prediction: $P(X_4 \mid \textbf{hot}_1, \textbf{cold}_2, \textbf{cold}_3)$

(We compute Part 4 before Part 3, since Part 3 depends on this result.)

Propagating $P(X_3 \mid e_{1:3}) = \langle 0, 0.2, 0.8, 0, 0, 0\rangle$ forward one step with no new evidence:

$$P(X_4 \mid e_{1:3}) = \sum_{X_3} P(X_4 \mid X_3)\, P(X_3 \mid e_{1:3})$$

$$\begin{aligned}
&P(X_4 = A) = 0 \quad \text{(unreachable)} \\
&P(X_4 = B) = P(B \mid B) \times 0.2 + P(B \mid C) \times 0.8 = 0.2 \times 0.2 + 0 \times 0.8 = 0.04 \\
&P(X_4 = C) = P(C \mid B) \times 0.2 + P(C \mid C) \times 0.8 = 0.8 \times 0.2 + 0.2 \times 0.8 = 0.32 \\
&P(X_4 = D) = P(D \mid B) \times 0.2 + P(D \mid C) \times 0.8 = 0 \times 0.2 + 0.8 \times 0.8 = 0.64 \\
&P(X_4 = E) = 0 \quad \text{(unreachable in one step from } B \text{ or } C\text{)} \\
&P(X_4 = F) = 0
\end{aligned}$$

X_4	$P(X_4 \mid \text{hot}_1, \text{cold}_2, \text{cold}_3)$
A	0.00
B	0.04
C	0.32
D	0.64
E	0.00
F	0.00

$$\boxed{P(X_4 \mid e_{1:3}) = \langle 0, 0.04, 0.32, 0.64, 0, 0\rangle}$$

On day 4, the rover is most likely at position D with probability 0.64.

3. Prediction: $P(\mathbf{hot}_4 \mid \mathbf{hot}_1, \mathbf{cold}_2, \mathbf{cold}_3)$

$$P(\text{hot}_4 \mid e_{1:3}) = \sum_{X_4} P(E_4 = \text{hot} \mid X_4)\, P(X_4 \mid e_{1:3})$$

Only $X_4 \in \{A, D\}$ contribute ($P(\text{hot} \mid X_4) = 1$ only for those), and from Part 4, $P(X_4 = D \mid e_{1:3}) = 0.64$ while $P(X_4 = A \mid e_{1:3}) = 0$:

$$= 0 \times 0.04 + 0 \times 0.32 + 1 \times 0.64 + 0 \times 0 + 0 \times 0 = \boxed{0.64}$$

There is a **64%** probability of observing a hot reading on day 4.

Bonus (10 points): $P(\mathbf{hot}_4, \mathbf{hot}_5, \mathbf{cold}_6 \mid \mathbf{hot}_1, \mathbf{cold}_2, \mathbf{cold}_3)$

We work backwards using the likelihood of future evidence:

$$P(e_{4:6} \mid e_{1:3}) = \sum_{X_3} P(X_3 \mid e_{1:3})\, P(e_{4:6} \mid X_3)$$

Step 1 — Compute $\beta_6(X_5) = P(E_6 = \mathbf{cold} \mid X_5)$:

$$\beta_6(X_5) = \sum_{X_6} P(X_6 \mid X_5)\, P(\text{cold} \mid X_6)$$

X_5	Computation	$\beta_6(X_5)$
A	$0.2 \times P(\text{cold} \mid A) + 0.8 \times P(\text{cold} \mid B) = 0.2 \times 0 + 0.8 \times 1$	0.8
B	$0.2 \times 1 + 0.8 \times 1$	1.0
C	$0.2 \times 1 + 0.8 \times 0$	0.2
D	$0.2 \times 0 + 0.8 \times 1$	0.8
E	$0.2 \times 1 + 0.8 \times 1$	1.0
F	1×1	1.0

$$\beta_6 = \langle 0.8, 1.0, 0.2, 0.8, 1.0, 1.0 \rangle$$

Step 2 — Compute $\beta_{56}(X_4) = P(E_5 = \mathbf{hot}, E_6 = \mathbf{cold} \mid X_4)$:

$$\beta_{56}(X_4) = \sum_{X_5} P(X_5 \mid X_4)\, P(\text{hot} \mid X_5)\, \beta_6(X_5)$$

X_4	Computation	$\beta_{56}(X_4)$
A	$0.2 \times P(\text{hot}\|A) \times 0.8 + 0.8 \times P(\text{hot}\|B) \times 1.0 = 0.2 \times 1 \times 0.8 + 0 = 0.16$	0.16
B	$0.2 \times 0 \times 1.0 + 0.8 \times 0 \times 0.2 = 0$	0
C	$0.2 \times 0 \times 0.2 + 0.8 \times P(\text{hot}\|D) \times 0.8 = 0 + 0.8 \times 1 \times 0.8 = 0.64$	0.64
D	$0.2 \times P(\text{hot}\|D) \times 0.8 + 0.8 \times 0 \times 1.0 = 0.2 \times 1 \times 0.8 + 0 = 0.16$	0.16
E	$0.2 \times 0 + 0.8 \times 0 = 0$	0
F	0	0

$$\beta_{56} = \langle 0.16, 0, 0.64, 0.16, 0, 0 \rangle$$

Step 3 — Compute $\beta_{456}(X_3) = P(E_4 = \textbf{hot}, E_5 = \textbf{hot}, E_6 = \textbf{cold} \mid X_3)$:

$$\beta_{456}(X_3) = \sum_{X_4} P(X_4 \mid X_3)\, P(\text{hot} \mid X_4)\, \beta_{56}(X_4)$$

X_3	Computation	$\beta_{456}(X_3)$
B	$0.2 \times P(\text{hot} \mid B) \times 0 + 0.8 \times P(\text{hot} \mid C) \times 0.64 = 0 + 0$	0
C	$0.2 \times P(\text{hot} \mid C) \times 0.64 + 0.8 \times P(\text{hot} \mid D) \times 0.16 = 0 + 0.8 \times 1 \times 0.16$	0.128

$$\beta_{456} = \langle 0, 0, 0.128, 0, 0, 0 \rangle \quad \text{(nonzero only at } X_3 = C\text{)}$$

Final combination:

$$\begin{aligned}
P(\text{hot}_4, \text{hot}_5, \text{cold}_6 \mid e_{1:3}) &= \sum_{X_3} P(X_3 \mid e_{1:3})\, \beta_{456}(X_3) \\
&= 0 \times 0 + 0.2 \times 0 + 0.8 \times 0.128 + 0 \times 0 + 0 \times 0 + 0 \times 0 \\
&= \boxed{0.1024}
\end{aligned}$$

The probability of observing hot, hot, cold on days 4, 5, and 6 is **10.24%**. This requires the rover to reach vent D by day 4 (probability 0.64), remain at D on day 5 (probability 0.2), and then move to E on day 6 (probability 0.8): $0.64 \times 0.2 \times 0.8 = 0.1024$.

5 Problem 5 (10 points): Markov Decision Process

States $\{A, B\}$, actions $\{1, 2\}$, discount $\gamma = 1$.

s	a	s'	$T(s, a, s')$
A	1	A	1.0
A	1	B	0.0
A	2	A	0.5
A	2	B	0.5
B	1	A	0.0
B	1	B	1.0
B	2	A	0.0
B	2	B	1.0

s	a	$R(s, a)$
A	1	0
A	2	-1
B	1	5
B	2	0

1. Bellman Equation

$$V^\pi(s) = R\big(s, \pi(s)\big) + \gamma \sum_{s'} T\big(s, \pi(s), s'\big)\, V^\pi(s')$$

With $\gamma = 1$:

$$\boxed{V^\pi(s) = R\big(s, \pi(s)\big) + \sum_{s'} T\big(s, \pi(s), s'\big)\, V^\pi(s')}$$

2. Two Iterations of Policy Evaluation ($\pi(A) = 1,\ \pi(B) = 1$)

Initialise $V_0^\pi(A) = V_0^\pi(B) = 0$.

Iteration 1:

$$V_1^\pi(A) = R(A, 1) + T(A, 1, A)\, V_0^\pi(A) + T(A, 1, B)\, V_0^\pi(B) = 0 + 1 \times 0 + 0 \times 0 = 0$$
$$V_1^\pi(B) = R(B, 1) + T(B, 1, A)\, V_0^\pi(A) + T(B, 1, B)\, V_0^\pi(B) = 5 + 0 \times 0 + 1 \times 0 = 5$$

Iteration 2:

$$V_2^\pi(A) = R(A, 1) + T(A, 1, A)\, V_1^\pi(A) + T(A, 1, B)\, V_1^\pi(B) = 0 + 1 \times 0 + 0 \times 5 = \boxed{0}$$
$$V_2^\pi(B) = R(B, 1) + T(B, 1, A)\, V_1^\pi(A) + T(B, 1, B)\, V_1^\pi(B) = 5 + 0 \times 0 + 1 \times 5 = \boxed{10}$$

3. Policy Improvement

$$\pi_{\text{new}}(s) = \arg\max_a \Big[R(s, a) + \sum_{s'} T(s, a, s')\, V_2^\pi(s')\Big]$$

State A:

$$\begin{aligned} a = 1: &\quad 0 + 1 \times 0 + 0 \times 10 = 0 \\ a = 2: &\quad -1 + 0.5 \times 0 + 0.5 \times 10 = 4 \end{aligned}$$

Since $4 > 0$: $\boxed{\pi_{\text{new}}(A) = 2}$

State B:

$$\begin{aligned} a = 1: &\quad 5 + 0 \times 0 + 1 \times 10 = 15 \\ a = 2: &\quad 0 + 0 \times 0 + 1 \times 10 = 10 \end{aligned}$$

Since $15 > 10$: $\boxed{\pi_{\text{new}}(B) = 1}$

Interpretation. The improved policy switches state A from action 1 to action 2. Staying at A forever (action 1) yields value 0, whereas action 2 gives a 50% chance of reaching B each step (which has value 10), making the expected return $-1 + 0.5 \times 10 = 4 > 0$. State B retains action 1, as it yields $5 + 10 = 15$ versus action 2's $0 + 10 = 10$.